

Mechanism of action of interleukin-13 antagonist (IL-13E13K) in cells expressing various types of IL-4R

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Received 11 December 2003; accepted 2 June 2004
Available online 29 July 2004

Abstract

As interleukin (IL)-13 and IL-4 play a major role in various diseases including asthma, allergy, and malignancies, it is desirable to generate a molecule that blocks the effects of both cytokines. We previously generated a human IL-13 mutant (IL-13E13K), which is a powerful antagonist of IL-13, blocking the biological activities of IL-13. We now show that IL-13E13K also competitively inhibits signaling and biological activities of IL-4 through type II and partially through type III IL-4 receptor (R) system. IL-13E13K completely blocked the IL-4-induced phosphorylation of STAT6 and IL-4-dependent protein synthesis in cells expressing type II and partially type III IL-4R but not type I IL-4R. Consistent with the inhibition of biological activities, IL-13E13K inhibited IL-4 binding to type II IL-4R-expressing cells but not to type I IL-4R-expressing cells. The inhibition efficiency of IL-4 binding by IL-13E13K was relatively lower compared to wtIL-13 even though IL-13E13K bound to IL-13R α 1 positive cells with a similar affinity to wtIL-13. These results indicate that Glu13 in IL-13 associates with IL-4R α , and mutation to lysine decreases its binding ability to IL-4R α chain. IL-13E13K binds to IL-13R α 1, which is shared by both IL-13R and IL-4R systems. Consequently, IL-13E13K inhibits IL-4 binding to these cells and prevents heterodimer formation between IL-13R α 1 and IL-4R α chains. This interference by IL-13E13K blocks the biological activities of not only IL-13 but also partially of IL-4. Thus, IL-13E13K may be a useful agent for the treatment of diseases such as asthma, allergic rhinitis, and cancer, which are dependent on signaling through both IL-4 and IL-13 receptors. Published by Elsevier Inc.

Keywords: Interleukin-4; Interleukin-13; Antagonist; Receptor; Signal transduction

1. Introduction

Interleukin-13 (IL-13) acts as a mediator or regulator of asthma, allergic and inflammatory diseases, and cancer [1,2]. This cytokine, predominantly produced by activated T-cells, induces IgE synthesis, expression of vascular cell adhesion molecule-1 (VCAM-1) on endothelial cells, MHC class II and CD23 in monocytes, and stimulates B-cell proliferation [3]. IL-13 also regulates TNF- α expression and the production of proinflammatory cytokines by monocytes [4]. IL-13 is known to be closely related to interleukin-4 (IL-4) func-

tionally and structurally although they share only 25% homology. IL-4 and IL-13 share many biological functions that are critical to the allergic and inflammatory responses, and cancer. However, unlike IL-4, IL-13 has no effects on T-cells [5].

Receptors for IL-4 (IL-4R) and IL-13 (IL-13R) share at least two components with each other [6–9] which are expressed in a variety of cell types [10,11]. The IL-4R system has been shown to be composed of IL-4R α (also known as IL-4R β), a primary IL-4-binding protein. This receptor forms a heterodimer with either IL-2R γ (common γ_c) chain (type I IL-4R) or IL-13R α 1 chain (type II IL-4R) for signaling. Cells that express all three chains (IL-4R α , IL-13R α 1, and γ_c chain) are termed here type III IL-4R. This type of IL-4R complex has been

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previously reported in several cell types [8,12,13], however, it is still not known whether all three chains bind IL-4 simultaneously or participate in IL-4 signaling.

IL-4R α chain binds to IL-4 with high affinity ($K_d = 20\text{--}300\text{ pM}$) and plays a critical role not only in IL-4 signaling but also IL-13 signaling [14–17]. In type I IL-4R complex, common γ_c chain forms complex with IL-4R α chain. It is hypothesized that after IL-4 binding to IL-4R α chain, the IL-4/IL-4R α complex recruits γ_c chain to form a heterodimer after IL-4 undergoes a conformational change allowing it to bind to γ_c chain [18,19]. IL-13R α_1 (also termed IL-13R α') chain binds IL-13 with moderate affinity (2–10 nM), however, it does not bind IL-4 directly [9,20]. IL-13R α_1 forms a complex with IL-4R α , and both chains are shared by the IL-4R and IL-13R systems. The complex has a higher affinity to IL-13 compared to each individual chain. The third component of the IL-13R system, IL-13R α_2 (also known as IL-13R α) chain, binds to IL-13 with higher affinity but does not bind IL-4 [21,22]. It has been shown that this receptor is a non-signaling component in IL-13R system and may serve as a “decoy” receptor. Whether this receptor is involved in IL-4R system is not clear [23].

In previous studies, various mutated proteins have been generated and utilized to clarify the mechanism of intimate ligand–receptor interaction in IL-4/IL-13 shared receptor systems. For example, an IL-4 mutant IL-4Y124D, which is a powerful antagonist of IL-4, was found to have cross-reactivity with both IL-4R and IL-13R systems [6]. Glu9 in IL-4 forms numerous hydrogen bonds to IL-4R α chain when IL-4/IL-4R α forms a complex, and it has been reported that IL-4E9K, in which Glu9 in IL-4 was substituted by Lys, was incapable of binding to IL-4R α -shared subunit, thus Glu9 in IL-4 is thought to be one of the important sites for binding to IL-4R α [24,25]. We previously reported a novel mutant of human IL-13, in which a glutamic acid (E) at position 13 was substituted by a lysine (K) residue [26]. This mutant (termed IL-13E13K)¹ acts as a powerful antagonist that inhibits biological activities of human IL-13 [26]. It binds with 4–8-fold higher affinity to IL-13R as compared to wild-type (wt) IL-13 with no detectable proliferative activity on TF-1 hematopoietic cells. IL-13E13K also inhibits IL-13-induced STAT6 activation in both immune and cancer cells and reverses IL-13-induced inhibition of CD14 expression on human primary monocytes.

In this study, we investigated whether IL-13E13K binds to IL-4R and affects IL-4 signaling and biological activities. We utilized various cell types that have been previously shown to express type I (Jurkat and Ramos), type II (THP-1 and COS-7), or type III (TF-1, Raji, and B-CLL) IL-4R. Our data indicate that IL-13E13K has

an impact on cells expressing type II IL-4R and some cells expressing type III IL-4R but not in cells that only express type I IL-4R complex.

2. Experimental procedures

2.1. Cell culture, reagents, and plasmids

Human Jurkat, Ramos, THP-1, PM-RCC, MDA-MB-231, TF-1, Raji, and monkey COS-7 cell lines were purchased from the American Type Culture Collection (Manassas, VA) and cultured as previously described [10]. Human primary peripheral blood mononuclear cells (PBMCs) were isolated from healthy adult volunteers by centrifugation over Ficoll–Hypaque gradient. The human primary B-chronic lymphocytic leukemia (B-CLL) cells were obtained from Dr. Neil Kay, Mayo Clinic, Rochester, MN. MDA-MB-231 cells were stably transfected with IL-13R α_2 or vector cDNA as described previously [27]. Recombinant human wtIL-13 and IL-4 were purchased from PeproTech (Rocky Hill, NJ). Recombinant IL-13E13K was produced and purified in our laboratory [34]. Recombinant IL-4 cytotoxin, IL4(38–37)-PE38K-DEL (termed IL-4PE), which contains a circularly permuted IL-4 mutant in which amino acids 38–129 were linked to amino acids 1–37 via a GGNGG linker and then fused to truncated *Pseudomonas* exotoxin, PE38K-DEL, consisting of amino acids 253–364 and 381–608 of PE followed by KDEL, was produced as described previously, and provided by Neurocrine Biosciences [28].

2.2. Protein synthesis assay

In the protein synthesis assay, Jurkat, THP-1, TF-1 (1×10^4), or human B-CLL (1×10^5) cells were incubated with different concentrations of human IL-4 in the presence or absence of IL-13E13K for 5 days at 37°C in 10% dialyzed FBS-containing leucine-free medium. Jurkat and THP-1 cells were cultured in serum-free medium. Then, 2 μCi [^3H]leucine (NEN Research Products, Boston, MA) was added to each well, and cells were incubated for an additional 8 h. The *in vitro* cytotoxic activity of IL-4PE was measured by the inhibition of protein synthesis [29]. All assays were performed in quadruplicate, and the concentration of IL-4 cytotoxin at which 50% inhibition of protein synthesis occurred was calculated (IC_{50}). The cytotoxic activity of IL-4 cytotoxin as measured by the inhibition of protein synthesis is directly proportional to cell death [29].

2.3. CD23 expression assay

Ramos B-cell lymphoma cells or human PBMCs were incubated for 24 or 48 h at 37°C with medium alone or different concentration of IL-4 (1 pg/ml–10 ng/ml) in the

¹ Abbreviations used: IL-13E13K, human interleukin-13E13K mutant; wtIL-13, wild-type human interleukin-13.

presence or absence of IL-13E13K (1 ng/ml–10 µg/ml). Cells were then stained with anti-CD23 Ab (BD Pharmingen, San Diego, CA) or IgG1 iso-type control with or without anti-CD19 Ab. Expression levels of CD23 were analyzed by FACSscan (Becton–Dickinson, San Jose, CA).

2.4. Immunoprecipitation and Western blotting

To detect the expression of IL-4R chains, cells (3×10^6) were washed by PBS and lysed at 4 °C with 50 mM Tris–HCl (pH 7.5)/150 mM NaCl/1% NP-40, containing proteinase inhibitors, Complete Mini (Roch Applied Science, Indianapolis, IN). The lysate was precleared by protein G–Sepharose beads (Santa Cruz, Santa Cruz, CA) and further incubated with protein G–Sepharose beads that had been preincubated with anti-IL-2R γ polyclonal Ab (C-20) or anti-IL-4R α polyclonal Ab (H-300, Santa Cruz) for 2 h at 4 °C. The resulting complex was washed five times in lysis buffer, resuspended in loading buffer (Invitrogen, Carlsbad, CA), and electrophoresed through 4–20% gradient SDS–PAGE. After electrophoresis, proteins were electrophoretically transferred onto PVDF membranes. The blotted membrane was blocked with PBS containing 5% skimmed milk at room temperature overnight, and then the membrane was reacted with anti-IL-2R γ monoclonal Ab (BD Biosciences Pharmingen) or anti-IL-4R α (H-300) for 2 h. Immunoreactive signal was visualized by enhanced chemiluminescence (ECL) substrate mixture according to the manufacturer's instructions (Amersham, Arlington Heights, IL). To detect the expression of IL-13R α 1 monoclonal Ab (R&D System, Minneapolis, MN) was used.

For analysis of inhibition of STAT6 phosphorylation by IL-13E13K, cells (2×10^6) were incubated with 50 ng/ml IL-4 in the presence or absence of IL-13E13K (1.25–10 µg/ml) for 15 min at 37 °C. The reactions were quenched by 10 ml of cold PBS containing 1 mM sodium vanadate, 25 mM NaF, 10 mM sodium pyrophosphate, and 1 mM EDTA. The cells were lysed at 4 °C with 1% Nonidet P-40, 300 mM NaCl, 50 mM Tris–HCl (pH 7.5), 10 µg/ml leupeptin, 10 µg/ml aprotinin, 1 mM PMSF, 1 mM sodium vanadate, 25 mM NaF, 10 mM sodium pyrophosphate, and 1 mM EDTA. The blotted membrane was blocked with PBS containing 5% skimmed milk at room temperature for 1 h, and then the membrane was reacted with anti-phosphorylated STAT6 polyclonal Ab (Cell Signaling Technology, Beverly, MA) overnight. Immunoreactive signal was visualized by ECL system. After stripping, the membrane was incubated with anti-STAT6 Ab (s-20, Santa Cruz) for detection of whole STAT6.

2.5. Radioreceptor binding assay

Recombinant human IL-4 and IL-13 were labeled with 125 I (Amersham) by the IODO-GEN iodination

reagent (Pierce, Rockford, IL) according to the manufacturer's instructions. The IL-4 or IL-13 equilibrium binding studies were performed by the previously described method [13]. Briefly, 1×10^6 cells in 100 µl binding buffer (RPMI 1640 containing 0.2% human serum albumin and 10 mM Hepes) were incubated for 2 h with 100–200 pM [125 I]IL-4 (specific activity, 16.7 µCi/µg) or [125 I]IL-13 (specific activity, 18.8 µCi/µg) with or without various concentrations (500 pM–40 nM) of unlabeled IL-4, wtIL-13, or IL-13E13K at 4 °C. Cell-bound cytokine was separated from unbound by centrifugation through a cushion of phthalate oils. Radioactivities in pelleted cells were counted by a gamma counter.

3. Results

3.1. Analysis of IL-4 receptor expression in various cell types

To analyze types of IL-4R expression in various cells, we examined the expression of each IL-4 receptor chain by Western blot analysis. As shown in Fig. 1, IL-4R α

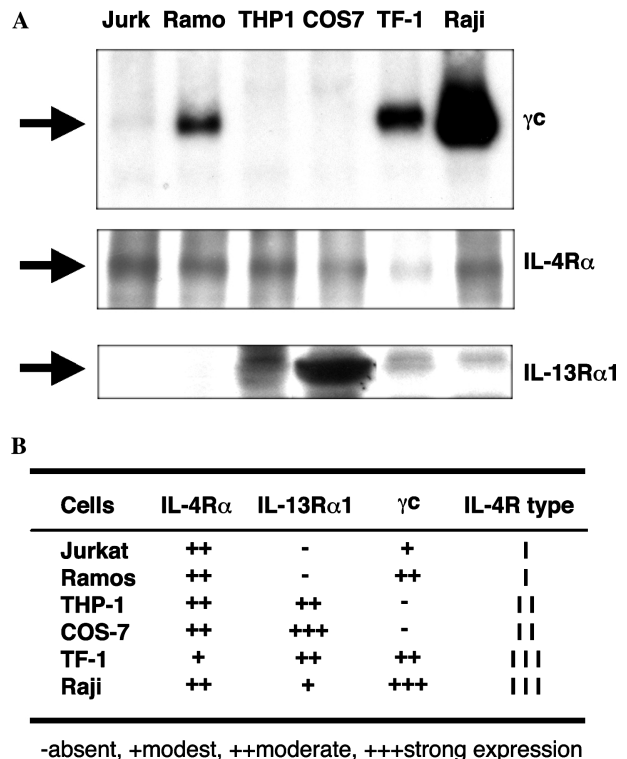


Fig. 1. Expression of IL-4R in various cells. (A) Cells (3×10^6) were washed by PBS and lysed at 4 °C. The lysate was immunoprecipitated with anti-IL-2R γ polyclonal Ab (C-20) or anti-IL-4R α polyclonal Ab (H-300). After the SDS–PAGE and blotting, membrane was reacted with anti-IL-2R γ monoclonal Ab or anti-IL-4R α . To detect the expression of IL-13R α 1 chain, Western blot was performed with anti-IL-13R α 1 monoclonal Ab. (B) Table summarizes the results of Western blot analysis.

chain is expressed in all cell lines tested. TF-1 cell line expressed lower level of IL-4R α chain. The expression of IL-13R α 1 chain varied. COS-7 cells expressed highest level of IL-13R α 1 chain followed by THP-1, TF-1, and Raji cells. However, Jurkat and Ramos cells did not express this receptor chain. The expression of γ_c chain was highest in Raji cells followed by TF-1, Ramos, and Jurkat cells. THP-1 and COS-7 cells did not express this receptor chain. Based on these findings, we classified Jurkat and Ramos as type I IL-4R-expressing cells, THP-1, and COS-7 as type II IL-4R-expressing cells, and TF-1 and Raji cells as type III IL-4R-expressing cells (Fig. 1B).

3.2. Effect of IL-13E13K on IL-4-induced protein synthesis in various cell types

Previously, we have reported that IL-13E13K is able to inhibit the stimulation of proliferation by wtIL-13 and IL-4 in TF-1 cells [26]. However, the effect of IL-13E13K on proliferation of other cell types is not known. This is because IL-4 cannot induce proliferation of various cell types. To address the issue whether IL-13E13K can inhibit the biological effect of IL-4 on various cell types, we determined whether IL-4 can induce protein

synthesis in various cell types expressing various IL-4R chains and whether IL-13E13K can inhibit their effect. For this purpose, we tested the effect of IL-4 and IL-13E13K on protein synthesis by the assessment of incorporation of [3 H]leucine in target cells. Although IL-4 induced protein synthesis in Jurkat cells (type I IL-4R-expressing cells), IL-13E13K did not inhibit this effect (Fig. 2A). However, in type II IL-4R-expressing cells (COS-7) IL-13E13K mediated strong effects and inhibited the IL-4-induced protein synthesis in a statistically significant and a dose-dependent manner (Fig. 2B). In cells expressing type III IL-4R such as TF-1 (Fig. 2C) and B-CLL (Fig. 2D) cells, IL-13E13K also inhibited IL-4-induced protein synthesis in a dose-dependent manner although the effects of IL-13E13K in B-CLL cells were moderate but statistically significant (Fig. 2E, $p < 0.05$). IL-13E13K also inhibited protein synthesis in TF-1 cells when stimulated by IL-13 (data not shown).

3.3. IL-13E13K inhibited CD23 expression induced by IL-4 in human primary B-lymphocytes but not in Ramos B-cell line

IL-4 induces CD23 expression on the surface of Ramos B-cell line [30]. To assess whether IL-13E13K can

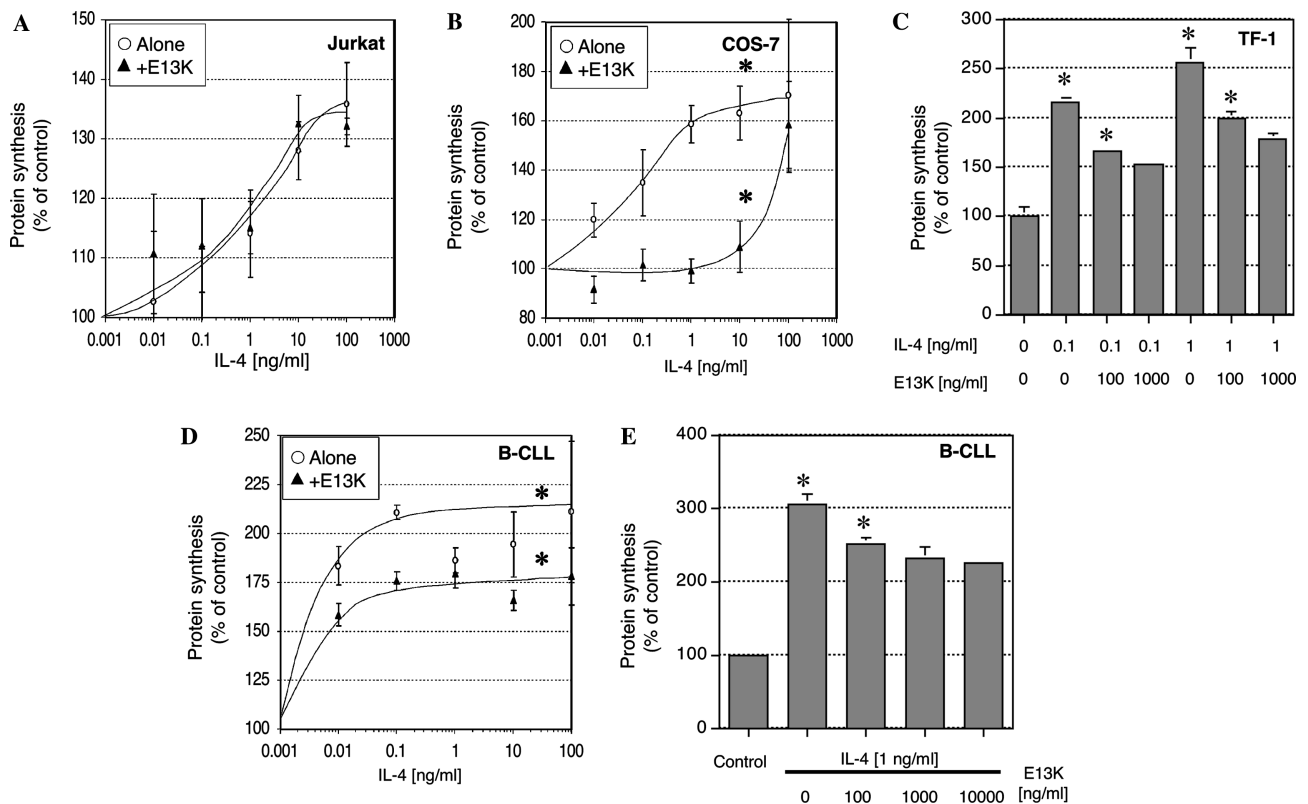


Fig. 2. IL-13 mutant inhibits the IL-4 or IL-13-stimulated protein synthesis in type II or type III IL-4R-expressing cells. (A,B,D) Cells were incubated with or without various concentrations of IL-4 (0.1–100 ng/ml) in the presence or absence of IL-13E13K (1000 ng/ml) for 5 days at 37°C. Cells were then pulsed with 2 μ Ci of [3 H]leucine for 8 h. (C,E) Cells were cultured in leucine-free medium containing 1 ng/ml of IL-4 with or without various concentrations of IL-13E13K (100–10,000 ng/ml) for 5 days. Cells were then pulsed with 2 μ Ci of [3 H]leucine for 8 h. Data shown are mean of quadruplicate determinations, and experiments were repeated several times. Bars, SD. *The difference is statistically significant at $p < 0.05$.

block the IL-4-induced CD23 expression, Ramos cells were incubated with IL-4 in the absence or presence of IL-13E13K. As shown in Fig. 3A (and data not shown), IL-4 induced CD23 expression in a dose-dependent manner in a concentration range from 0.1 to 100 ng/ml.

However, IL-13E13K did not inhibit the induction of CD23 despite the use of 1000–10,000 times excess amounts of IL-13E13K. Interestingly, when primary B-cells were used, IL-13E13K partially inhibited IL-4-induced CD23 expression in a dose-dependent-manner

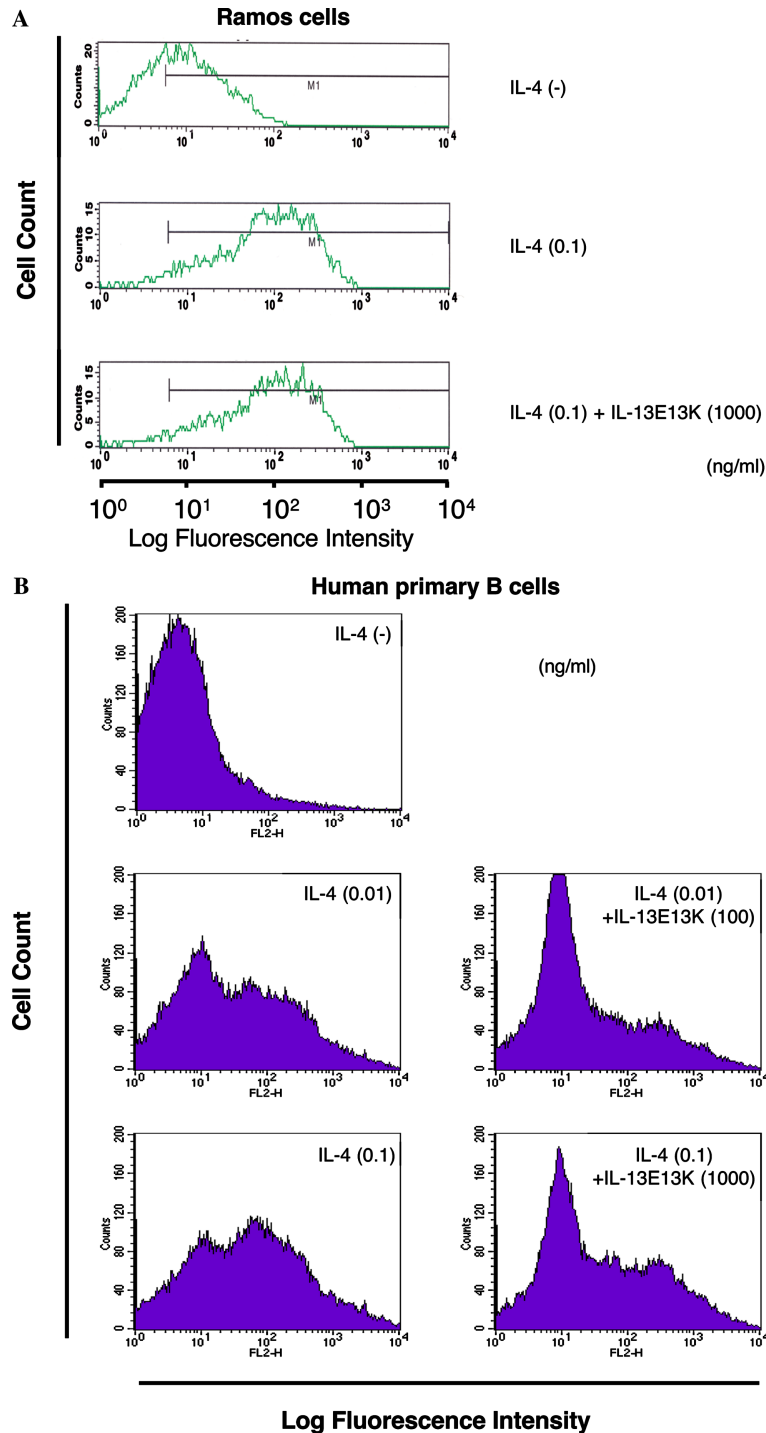


Fig. 3. IL-13E13K mediates inhibitory effects on IL-4-induced CD23 expression in human primary B-cells but not in Ramos B-cell line. (A) Ramos cells were cultured with or without 0.1 ng/ml of IL-4 in the presence or absence of IL-13E13K (1000 ng/ml) for 48 h. (B) Human primary B-cells were cultured with or without 0.01 and 0.1 ng/ml IL-4 with 100 and 1000 ng/ml IL-13E13K for 24 h. Both cell types were then washed and stained as described in Experimental procedures.

(Fig. 3B). As primary B-cells have been vigorously studied and shown to express type III IL-4R [1,2], our results indicate that IL-13E13K can inhibit the biological activity of IL-4 in cells expressing all three chains (IL-4R α , IL-13R α 1, and γ_c) of IL-4Rs but not in cells that only express two chains (IL-4R α and γ_c).

3.4. IL-13E13K inhibits signal transduction induced by IL-4 in type II and type III IL-4R-expressing cells

We previously demonstrated that IL-13E13K does not stimulate STAT6 activation and inhibits wtIL-13-induced STAT6 activation in THP-1 cells [26]. We now examined whether IL-13E13K can inhibit IL-4-induced signaling through STAT6 pathway in various types of IL-4R-expressing cells. As shown in Fig. 4, IL-4-induced STAT6 phosphorylation in Jurkat, Ramos, THP-1, COS-7, TF-1, and Raji cells as also reported by others [26,31–33]. Consistent with protein synthesis assays, IL-13E13K did not inhibit IL-4-induced STAT6 phosphorylation in type I IL-4R-expressing cells, but it did inhibit IL-4-induced STAT6 phosphorylation when 25- to 200-fold excess of IL-13E13K was used in THP-1 and COS-7 cells that express type II IL-4R. At 200-fold excess of IL-13E13K, a complete inhibition of IL-4-in-

duced STAT6 phosphorylation was observed. Interestingly, IL-13E13K partially inhibited IL-4-induced STAT6 phosphorylation in TF-1 cells but not in Raji cells despite both cell types express type III IL-4R.

3.5. IL-13E13K neutralizes cytotoxic activity of IL-4 cytotoxin in COS-7 and TF-1 cells but not in Jurkat and Raji cells

IL-4 cytotoxin (IL4-PE) targets IL-4R on cancer cell surfaces, and internalization of this molecule into cells mediates inhibition of protein synthesis, subsequently causing cell death [28]. This cytotoxin is found to be highly cytotoxic to IL-4R positive tumor cells in vitro and in vivo [34]. To assess the interaction between IL-13E13K and IL-4/IL-4R, we used type I IL-4R-positive cells (Jurkat), type II IL-4R-positive cells (COS-7), and type III IL-4R-positive cells (TF-1 and Raji) and determined the effect of IL-13E13K on IL-4 cytotoxin-induced cytotoxicity. The cytotoxicity was assessed by the measurement of inhibition of protein synthesis. As shown in Fig. 5, IL4-PE was modestly cytotoxic to Jurkat and TF-1 cells but highly cytotoxic to COS-7 and Raji cells. As excess of IL-4 neutralized the cytotoxic effects of IL4-PE in all tested cells, it was concluded that

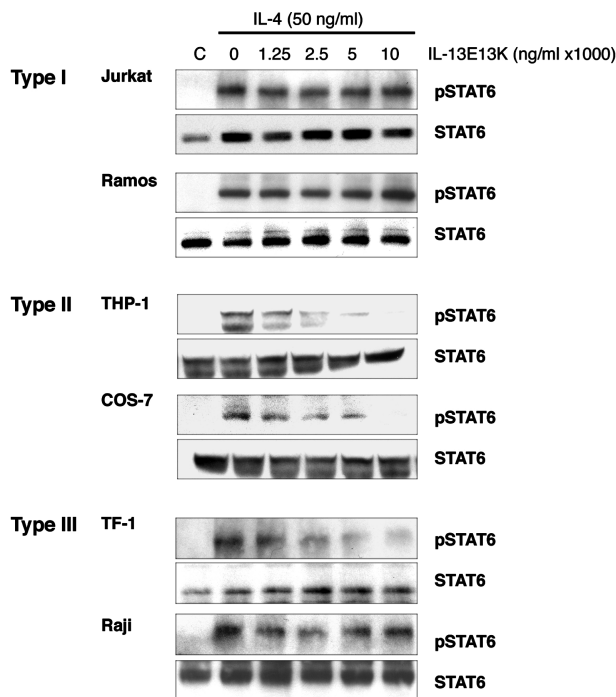


Fig. 4. Inhibition of IL-4-stimulated STAT6 phosphorylation by IL-13E13K in THP-1, COS-7, and TF-1 cells. Jurkat, Ramos, THP-1, COS-7, TF-1, or Raji cells were incubated with or without 0.05 μ g/ml IL-4 in the presence or absence of IL-13E13K (1.25, 2.5, 5, or 10 μ g/ml) for 15 min. Cell lysates (2×10^6 cells/sample) were separated by 4–20% gradient SDS-PAGE, transferred to PVDF membranes, and immunoblotted with 4G10 Ab, which was visualized by ECL detection kit. pSTAT6 represents phosphorylated STAT6.

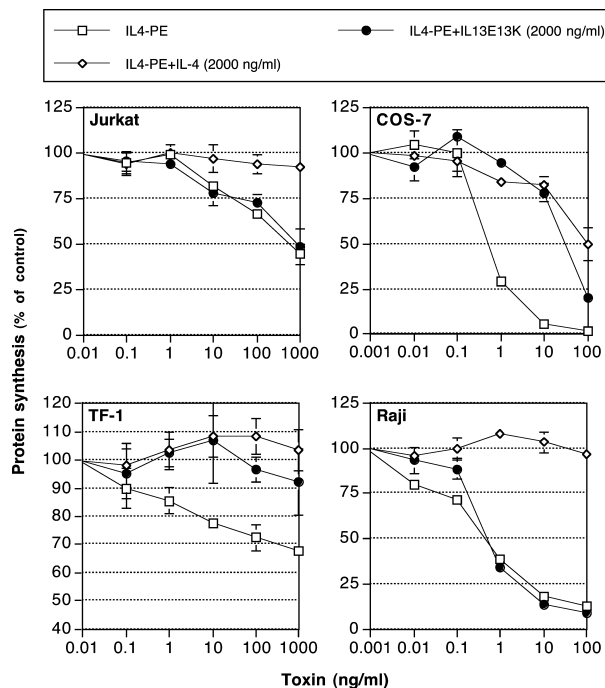


Fig. 5. IL-13E13K neutralizes the cytotoxic activity of IL-4PE in COS-7 and TF-1 cells, but not in Jurkat and Raji cells. Cells (5×10^3 /well Jurkat cells and 1×10^4 /well others) were cultured in leucine-free medium containing various concentrations of IL-4 cytotoxin (IL4-PE) with or without 2 μ g/ml of IL-13E13K or IL-4 for 20 h. Cells were then pulsed with 1 μ Ci of [3 H]leucine for 4 h. Data shown are mean of quadruplicate determinations, error bars representing SD. Experiments were repeated several times.

this cytotoxin exerts its cytotoxicity specifically through IL-4R. Interestingly, IL-13E13K was also able to neutralize IL-4 cytotoxin-induced cytotoxicity in COS-7 cells, and the IC_{50} was altered from 0.5 to 20 ng/ml. In contrast, IL-4 cytotoxin was highly or modestly cytotoxic to Raji or Jurkat cells (IC_{50} = 0.9 and 600 ng/ml, respectively), however, IL-13E13K failed to neutralize the effect of IL-4 cytotoxin in both cell types. On the other hand, IL-13E13K clearly neutralized the cytotoxicity of IL4-PE on TF-1 cells although the cytotoxicity of IL4-PE in these cells was very low. These results indicate that IL-13E13K neutralizes the cytotoxic activity of IL-4 cytotoxin in cells expressing type II IL-4R, whereas it does not neutralize the cytotoxicity in cells expressing type I IL-4R. A mixed result was observed in cells ex-

pressing type III IL-4R (TF-1 and Raji). IL-13E13K did not neutralize the cytotoxic effect of IL-4 cytotoxin in Raji cells but it did in TF-1 cells. These results suggest that IL-13E13K either inhibits IL-4 binding on the cells or internalization of cell-bound IL-4.

3.6. IL-13E13K competes for the binding of radiolabeled IL-4

Since we found that IL-13E13K was able to inhibit signal transduction through IL-4R, we next examined the ability of IL-13E13K in competing [125 I]IL-4-binding to cells expressing either type II or type III IL-4R. As in most assays IL-13E13K did not inhibit IL-4 effect in type I IL-4R-expressing cells, these cells were not tested in binding

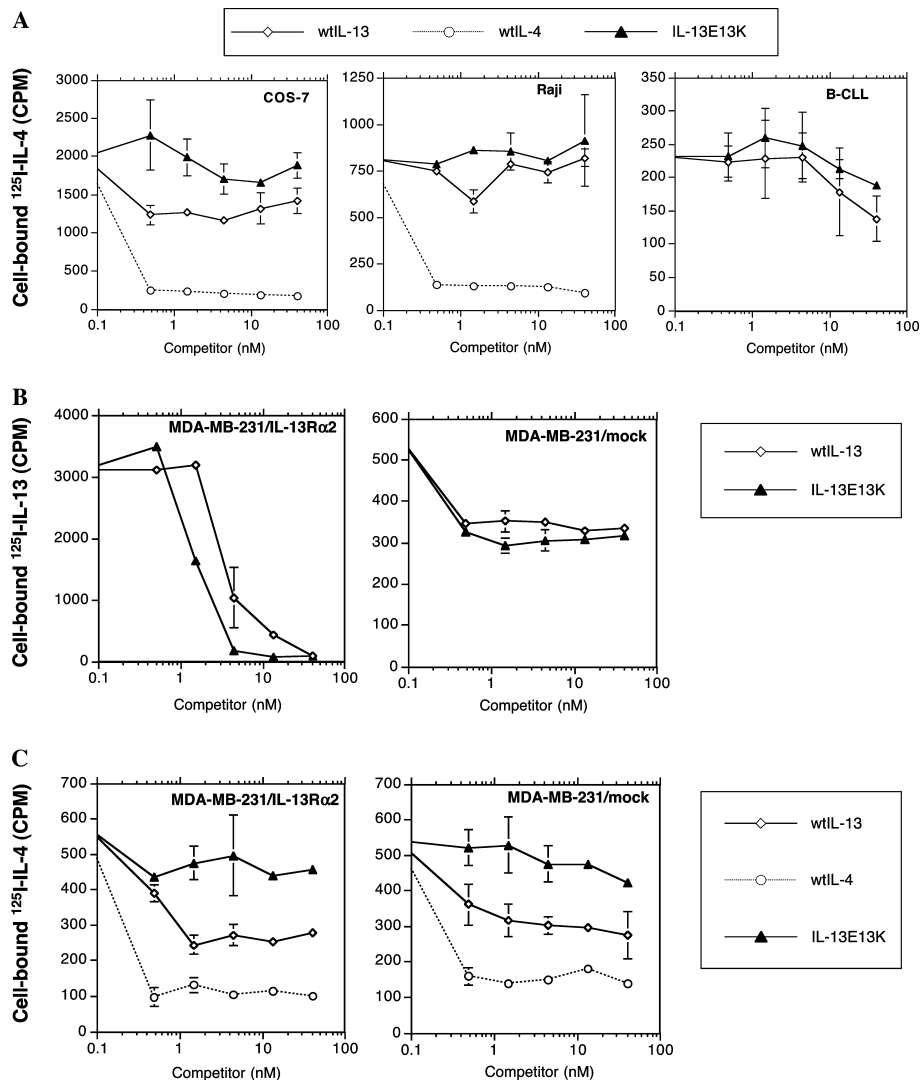


Fig. 6. Competitive ability of IL-13E13K in [125 I]IL-4 binding to type II positive and type III IL-4R positive cells. Cells (1×10^6) were incubated with [125 I]IL-4 (A) 200 pM and (C) 100 pM or [125 I]IL-13 (200 or 500 pM) (B) with or without increasing concentrations of unlabeled IL-4, wtIL-13, or IL-13E13K. Data are presented as mean binding (cpm) $\pm$ SD. Total [125 I]IL-4 bound (A,C) or [125 I]IL-13 (B) was 1987 ± 251 cpm in COS-7 cells, 818 ± 85 cpm in Raji cells, 240 ± 11 cpm in B-CLL cells (A), 3113 ± 227 cpm in MDA-MB-231 cells transfected with human IL-13R α 2 (MDA-MB-231/hIL-13R α 2), 573 ± 48 cpm in MDA-MB231 cells transfected with vector only (MDA-MB-231/mock) (B), 587 ± 92 cpm in MDA-MB-231/hIL-13R α 2 cells, and 541 ± 91 cpm in MDA-MB-231/mock cells (C).

assays. As shown in Fig. 6A, IL-13E13K modestly displaced [125 I]IL-4 binding in type II IL-4R-expressing COS-7. Interestingly, wtIL-13 inhibited the binding of [125 I]IL-4 strongly compared to IL-13E13K. Similar to other assays, type III IL-4R-expressing cells showed mixed results. Both wtIL-13 and IL-13E13K modestly displaced the binding of [125 I]IL-4 binding in B-CLL cells but not in Raji cells. These results demonstrate that IL-13E13K can modestly compete for the binding of IL-4 in type II and type III IL-4R-positive cells.

3.7. Impact of IL-13R α 2 expression on IL-13E13K binding to IL-4R

We have previously demonstrated that IL-13E13K possess 4- to 8-fold higher binding avidity compared to wtIL-13 in displacing [125 I]IL-13 binding to U251 and PM-RCC cells, which naturally express IL-13R α 2 chain [26]. U251 and PM-RCC cells express type II IL-4R [10]. However, the significance of IL-13R α 2 chain in the IL-4R and IL-13R systems when IL-13E13K binds to cells expressing this component is unclear. To determine the contribution of IL-13R α 2 chain when cells bind to IL-13E13K, we used MDA-MB-231 human breast cancer cell line stably transfected with IL-13R α 2 chain (MDA-MB-231/IL-13R α 2) to perform [125 I]IL-13 and [125 I]IL-4 binding studies. As shown in Fig. 6B, IL-13E13K was superior to wtIL-13 in displacing [125 I]IL-13 binding to MDA-MB-231/IL-13R α 2 cells whereas in MDA-MB-231/mock control cells both wtIL-13 and IL-13E13K displaced [125 I]IL-13 binding but no significant difference between each other was observed. To assess whether the presence of IL-13R α 2 chain also affects the competitive ability of IL-13E13K in [125 I]IL-4 binding, we performed binding assays in MDA-MB-231/IL-13R α 2 and MDA-MB-231/mock control cells. MDA-MB-231 cells express type II IL-4R [35]. As shown in Fig. 6C, both wtIL-13 and IL-13E13K displaced binding of [125 I]IL-4 to MDA-MB-231/IL-13R α 2 cells. In sharp contrast to [125 I]IL-13 binding, wtIL-13 was superior to IL-13E13K in competing for [125 I]IL-4 binding to these cells. There was no difference between IL-13R α 2-transfected cells and mock control cells in the ability of IL-13E13K to compete for [125 I]IL-4 binding. These results indicate that IL-13R α 2 chain does not influence in [125 I]IL-4 binding, and IL-13E13K has modest effect in competing for type II IL-4R.

4. Discussion

The major goal of our current study was to investigate whether IL-13 antagonist, IL-13E13K, can inhibit the signal transduction and biological activity of IL-4 and to analyze the interaction of IL-13E13K with the IL-4R and IL-13R systems. We demonstrate that IL-

13E13K can completely inhibit signaling and biological activities of IL-4 in type II IL-4R-expressing cells and partially in type III IL-4R-expressing cells. However, IL-13E13K did not inhibit signaling or biological activity of IL-4 in cells expressing type I IL-4R. Similar to the inhibition of IL-4 biological activity, IL-13E13K modestly competed for the binding of IL-4 in type II and type III IL-4R positive cells.

We confirmed the structure of IL-4R in various cell types by Western blot analysis and demonstrated that Jurkat and Ramos cells express type I IL-4R (consisting of IL-4R α and γ_c chains), THP-1 and COS-7 cells express type II IL-4R (consisting of IL-4R α and IL-13R α 1 chains), and TF-1 and Raji cells express type III (consisting of IL-4R α , γ_c , and IL-13R α 1 chains). These results confirm various previous reports demonstrating mRNA expression of these receptor chains in various cell types and our own RT-PCR studies [10,13,18]. Although B-CLL cells were not examined for the expression of various IL-4R chains in this study, previous study has demonstrated that these cells express type III IL-4R (Kay NE, et al. unpublished data). The results in the current study indicate that IL-13E13K only interacts with IL-4R complex if IL-13R α 1 chain is present. We also observed that IL-13E13K does not inhibit the IL-4-induced cell proliferation in CTLL-2 cells transfected human IL-4R α (data not shown). Taken together, our results indicate that IL-13E13K neither binds to IL-4R α chain alone nor affects the heterodimer formation between IL-4R α and γ_c chain. As summarized in Fig. 7, IL-13E13K is able to block IL-4 binding and consequently inhibit biological activities of IL-4 in cells expressing type II or type III IL-4R.

Although IL-13E13K caused mild to moderate displacement of radiolabeled IL-4 binding to COS-7 and TF-1 cells, this molecule clearly inhibited IL-4-induced protein synthesis or STAT6 activation in these cells. The possible reason for this is that IL-13E13K interferes in heterodimer formation of IL-4R α and IL-13R α 1 chains. IL-4R α chain does not bind IL-13 without the presence of IL-13R α 1, however, the IL-13R α 1/IL-4R α complex plays a crucial role in IL-13 signaling. Once IL-4R α forms a complex with IL-13R α 1, the complex has higher affinity with IL-13 compared to each individual chain. Based on these observations, a sequential-binding mechanism for IL-13 interaction to type I IL-13R has been proposed: IL-13 binds to IL-13R α 1, and then the complex recruits IL-4R α chain to make a heterodimer, or alternatively, IL-4R α /IL-4 complex recruits IL-13R α 1 chain to form a signaling heterodimer [36,37]. Although the majority of IL-4 binds IL-4R α chain in the presence of IL-13E13K, the IL-4/IL-4R α complex cannot recruit IL-13R α 1 chain, which results in failure to form IL-4R α /IL-13R α 1 heterodimer, as IL-13R α 1 chain is occupied with IL-13E13K. Alternatively, Glu at position 13 in

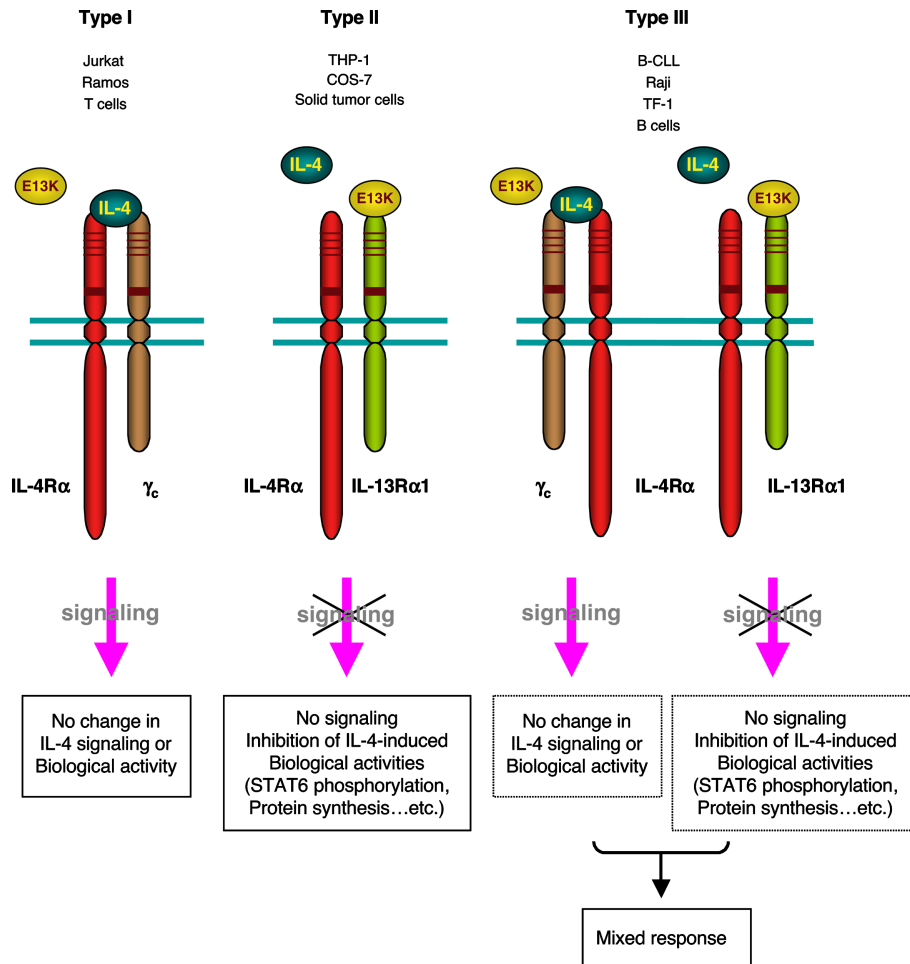


Fig. 7. Postulated schematic model for IL-13E13K interaction with IL-4R. IL-13E13K cannot compete for [125 I]IL-4 binding, IL-4 signaling, or biological activity of IL-4 in type I IL-4R-expressing cells. In contrast, IL-13E13K can efficiently inhibit the ligand binding, IL-4 signaling, and biological activity of IL-4 in cells expressing type II IL-4R. In type III IL-4R-expressing cells, the blocking ability of this mutant was variable; IL-13E13K could inhibit the biological activities of IL-4 in some cell types (TF-1, B-CLL, and primary B-cells) but not in all cell lines.

IL-13 is responsible for binding to IL-4Rα in the IL-13Rα1/IL-4Rα complex when it binds to IL-13.

IL-4-PE has been shown to mediate cytotoxicity in cells that express large numbers of IL-4R [28,34]. As expected, IL-4-PE mediated cytotoxicity in cells expressing all three types of IL-4R, and IL-13E13K neutralized cytotoxicity in type II or both type III IL-4R-positive cells. However, it did not neutralize cytotoxicity in type I IL-4R-expressing Jurkat cells. In TF-1 cells, IL-4-PE was modestly cytotoxic but IL-13E13K neutralized its cytotoxicity. These results are consistent with other effects of IL-13E13K on various cell types. It is interesting to note that Raji cells express type III IL-4R, however, this cell type responds differently from TF-1 or B-CLL cells which also express type III IL-4R. For example, (1) IL-13E13K does not inhibit IL-4-induced phosphorylation of STAT6 in Raji cells, but it does in TF-1 cells, (2) IL-13E13K does not neutralize IL-4-PE-induced cytotoxicity in Raji cells, but it does in TF-1 cells, and finally (3) IL-13E13K does not inhibit binding of [125 I]IL-4 in Raji cells. As Raji cells express ex-

cess of γc chain than any other cell type, it is possible that IL-4Rα is completely occupied in forming a heterodimer with γc, and no IL-4Rα chain is available to form a complex with IL-13Rα1 chain. Alternatively, Raji cells express a mutant form of IL-13Rα1 chain that does not allow binding of IL-13E13K.

Because IL-13E13K binds IL-13Rα1 positive and IL-13Rα2 negative cells (e.g., MDA-MB-231/mock) with a similar affinity to wtIL-13, Glu13 in IL-13 does not seem to associate with IL-13Rα1 chain directly. In the case of IL-4, it is reported that when IL-4 binds to IL-4Rα chain, Glu9 in IL-4 forms hydrogen bonds to IL-4Rα chain [25]. It is also reported that IL-4E9K cannot bind to IL-4Rα chain [24]. Since Glu13 is conserved in IL-13, it might bind to IL-4Rα chain. These possibilities support our previous hypothesis that the receptor binding interface of the IL-13 molecule is located in α-helix A and D, where α-helix D interacts with IL-13Rα1 subunit and α-helix A interacts with IL-4Rα subunit [26]. We found that IL-13E13K has higher binding affinity to MDA-MB-231 cells

transfected with IL-13R α 2 chain compared to wtIL-13 (Fig. 6B). This is probably because IL-13E13K is incapable of binding to IL-4R α chain resulting into increased binding of IL-13E13K to IL-13R α 2 chain. Nevertheless, inhibition of [¹²⁵I]IL-4 binding by IL-13E13K was not affected regardless of the presence or absence of IL-13R α 2 chain on the cell surface.

In conclusion, IL-13E13K inhibits the biological activities of not only IL-13 but also partially of IL-4. This property of IL-13E13K should be further exploited in the diseases (e.g., cancer and allergic asthma) in which both IL-4 and IL-13 play a major role in the pathogenesis. As the effect of IL-13E13K on IL-4 suppression is moderate, combinational therapeutic strategies in which IL-4 antagonist is combined with IL-13 antagonist may should be explored. Since IL-13 has been shown to be a key regulator of allergen-induced airway inflammation independent of IL-4 in a mouse model [38], IL-13E13K may offer an attractive option in the management of these diseases.

Acknowledgments

We are grateful to Ms. Pamela Dover for general help, technical support, and reading the manuscript and Dr. Hidekazu Shirota for helpful suggestion and technical support to purify PBMC and perform FACS analysis. We also thank Drs. Syed R. Husain and Bharat H. Joshi for helpful suggestions and Raymond P. Donnelly for critical reading the manuscript. The views presented in this article do not necessarily reflect those of the Food and Drug Administration.

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